

A11 SOURCE SEPARATION | O (MAX : 1 PT)

Intent : Preserve indoor air quality and maximize olfactory comfort in occupied spaces through the isolation and proper ventilation of indoor pollution sources and chemical storage areas.

Summary : This WELL feature requires strategies that isolate key sources of odors, germs, pollution or humidity through doors or dedicated exhaust.

Issue : Air pollution can be created from many indoor sources, including cleaning products, office equipment and humid environments. Chemical storage closets that house cleaning products, can be a source of harmful vapors, including Volatile Organic Compounds (VOCs) that are linked to cancer, organ and central nervous system damage.^{1,2} Copy rooms can contribute to the production of ozone, which has been associated with lung inflammation, chest pain, wheezing, coughing, shortness of breath and the exacerbation of respiratory illnesses, such as pneumonia and asthma.³⁻⁵ Bathrooms can encourage the growth of mold and mildew, which have been associated with allergic reactions and aggravate lung diseases, such as asthma.⁶ Exposure to air pollutants in kitchens, such as particulate matter, VOCs and polycyclic aromatic hydrocarbons, has been associated with kidney inflammation.⁷

Solutions : The most effective way to curb air pollution in buildings is to eliminate individual sources or capture emissions at the source, before they spread to surroundings.⁸ For air pollution sources that are inevitable, physical separation of such sources combined with direct ventilation exhaust systems is an effective means of protecting individuals.

PART 1 MANAGE POLLUTION AND EXHAUST (MAX : 1 PT)

For All Spaces except Dwelling Units & Guest Rooms:

For all bathrooms, kitchens, rooms for cleaning and chemical storage, rooms with high-volume printers and copiers and high-humidity areas, the following requirements are met:

- a. Meet one of the following:
 1. Are separated from all adjacent regularly occupied spaces with self-closing doors and/or vestibules.
 2. Are negatively pressurized compared with adjacent regularly occupied spaces.
- b. Utilize exhaust fans such that the return air is vented outdoors and not recirculated.^{9,10}

For Commercial Kitchen Spaces:

Option 1: Control cooking pollution

The following requirements are met:

- a. Canopy hoods have side or partial panels, when allowable by code.¹¹
- b. Type II hood overhangs and setbacks comply with ASHRAE 154-2011.¹²
- c. The vertical distance between the front lower lip of the hood and the cooking surface is less than or equal to 1.2 m.¹²
- d. Replacement air velocity near (or directed at) the hood is less than 0.38 m/s.¹¹
- e. Replacement air introduced directly into the exhaust hood cavity does not exceed 10% of the hood exhaust airflow rate.¹³
- f. Appliances are grouped under exhaust hoods according to effluent production and associated ventilation requirements, as specified in ASHRAE 154-2011, per hood type (defined by the classifications used in ASHRAE 154-2011 for light, medium, heavy and extra-heavy appliance duty levels).¹²
- g. Appliances have a rear seal between the appliance and the wall, when allowable by code.¹⁵
- h. Appliances located at the end of a cook line requiring exhaust airflow rates greater than 460 L/s/m have a full side panel or an end wall.¹²

OR

Option 2: No applicable commercial appliances

The following requirements are met:

- a. There are no appliances that require a hood, as specified in ASHRAE 154-2011 (Table 2 - Type I Hood Requirements by Appliance Type, Table 2 - Type II Hood Requirements by Appliance Description).¹²

For Dwelling Units & Guest Rooms:

The following requirements are met:

- a. Where cooking burners and stove top cooking appliances are present, a range hood meets the following:
 1. Exhaust air is vented directly to the outdoors.¹⁶
 2. Exhaust air outlets are separated from any air intakes by at least 3 m, unless otherwise specified by local code.¹⁷
 3. The minimum operating exhaust airflow rate is the greater of 150 L/s per linear meter of range hood width or 94 L/s.¹⁸
 4. The range hood device, when in operation, covers at least 75% of the burner area.¹⁹
- b. All bathrooms contain one of the following:

1. Exhaust fan.
2. Operable window.

REFERENCES

1. Nurmatov UB, Tagiyeva N, Semple S, Devereux G, Sheikh A. Volatile organic compounds and risk of asthma and allergy: A systematic review. *Eur Respir Rev.* 2015;24(135):92-101. doi:10.1183/09059180.00000714
2. U.S. Environmental Protection Agency. Volatile Organic Compounds' Impact on Indoor Air Quality. <https://www.epa.gov/indoor-air-quality-iaq/volatile-organic-compounds-impact-indoor-air-quality>.
3. U.S. Environmental Protection Agency. Health Effects of Ozone in Patients with Asthma and Other Chronic Respiratory Disease. <https://www.epa.gov/ozone-pollution-and-your-patients-health/health-effects-ozone-patients-asthma-and-other-chronic>.
4. The National Institute of Environmental Health Sciences. Ozone. <https://www.niehs.nih.gov/health/topics/agents/ozone/index.cfm>. Published 2008.
5. Tuomi T, Engstrom B, Niemela R, Svinhufvud J, Reijula K. Emission of ozone and organic volatiles from a selection of laser printers and photocopiers. *Appl Occup Environ Hyg.* 2000;15(8):629-634. doi:10.1080/10473220050075635
6. Mendell MJ, Mirer AG, Cheung K, Tong M, Douwes J. Respiratory and allergic health effects of dampness, mold, and dampness-related agents: A review of the epidemiologic evidence. *Environ Health Perspect.* 2011;119(6):748-756. doi:10.1289/ehp.1002410
7. Singh A, Kamal R, Mudiam MKR, et al. Heat and PAHs emissions in indoor kitchen air and its impact on kidney dysfunctions among kitchen workers in Lucknow, North India. Jia Z, ed. *PLoS One.* 2016;11(2):e0148641. doi:10.1371/journal.pone.0148641
8. U.S. Environmental Protection Agency. Improving Indoor Air Quality. <https://www.epa.gov/indoor-air-quality-iaq/improving-indoor-air-quality>.
9. U.S. Green Building Council. LEED BD+C: New Construction | v4 - LEED v4. 2018. https://www.usgbc.org/sites/default/files/LEED_v4_BDC_07.2.18_current.pdf.
10. Wenger Corporation. Planning your athletic facility. <https://www.wengercorp.com/Construct/docs/AthleticPlanningGuidebyWengerGearBoss.pdf>.
11. California Energy Commission. Design Guide Improving Commercial Kitchen Ventilation System Performance.; 2003. http://www.energy.ca.gov/reports/2003-06-13_500-03-034F.PDF.
12. International Codes Council. 2015 Minnesota Mechanical and Fuel Gas Code with the ANSI/ASHRAE Standard 154-2011. Presented at the: 2015. <https://codes.iccsafe.org/public/document/MMFGC2015/copyright>.
13. American Society of Heating Refrigerating and Air-Conditioning Engineers. ASHRAE 90.1: Energy Standard for Buildings Except Low-Rise Residential Buildings.; 2010. http://www.usalighting.com/stuff/contentmgr/files/1/b90ce247855d0f17438484c003877338/misc/ashrae_90_1_2010.pdf.
14. American Society of Heating Refrigerating and Air-Conditioning Engineers. ASHRAE 189.1: Standard for the Design of High-Performance Green Buildings Except Low-Rise Residential Buildings.; 2011. http://www.ditar.cl/archivos/Normas_ASHRAE/T1160ASHRAE-189.1-2009-Design-H-P-Green-Buildings.pdf.
15. Sobiski P, Swierczyna R, Fisher D. Effects of Range Top Diversity, Range Accessories, and Hood Dimensions on Commercial Kitchen Hood Performance. Vol 112 PART 1. San Francisco, CA; 2006.
16. Militello-Hourigan RE, Miller SL. The impacts of cooking and an assessment of indoor air quality in Colorado passive and tightly constructed homes. *Build Environ.* 2018. doi:10.1016/j.buildenv.2018.08.044
17. American Society of Heating Refrigerating and Air Conditioning Engineers (ASHRAE). ASHRAE 154-2003: Ventilation for Commercial Cooking Operations. 2003. http://sspc154.ashraepcs.org/pdf/STD154_2003.pdf.
18. Home Ventilating Institute. What is a range hood? https://www.hvi.org/HVIORG/document-server/?cfp=HVIORG/assets/File/public/publications/HVIRangeHood_4Feb08.pdf.
19. Singer BC, Delp WW, Apte MG. Experimental evaluation of installed cooking exhaust fan performance. Ernest Orlando Lawrence Berkeley Natl Lab. 2012.